

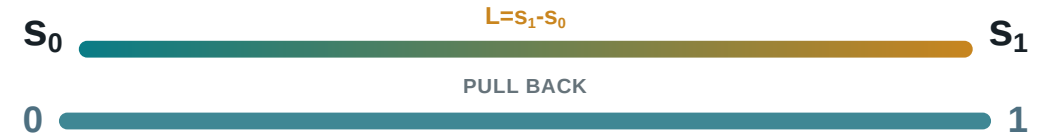
Unit-Interval Pull-Back and Operator Scaling

Physical length is absorbed into normalized operator coefficient profiles and source profile

PHYSICAL ONE-DIMENSIONAL AXIAL OPERATOR

$$s \in [s_0, s_1],$$
$$\mathcal{L}_{\text{phys}} u = -\frac{d}{ds} \left(a_{\text{phys}}(s) \frac{du}{ds} \right) + b_{\parallel, \text{phys}}(s) \frac{du}{ds} + c_{\text{phys}}(s) u = f_{\text{phys}}(s).$$

NORMALIZE THE COORDINATE



$$s = s_0 + Lt, \quad v(t) = u(s_0 + Lt), \quad t \in [0, 1].$$

The interval is normalized, but the physical length is not discarded.

$$a_{\text{unit}} = a_{\text{phys}}, \quad a'_{\text{unit}} = La'_{\text{phys}}, \quad b_{\parallel, \text{unit}} = Lb_{\parallel, \text{phys}}, \quad c_{\text{unit}} = L^2 c_{\text{phys}}, \quad f_{\text{unit}} = L^2 f_{\text{phys}}.$$

$$-\frac{d}{dt} \left(a_{\text{unit}}(t) \frac{dv}{dt} \right) + b_{\parallel, \text{unit}}(t) \frac{dv}{dt} + c_{\text{unit}}(t) v(t) = f_{\text{unit}}(t), \quad t \in [0, 1].$$

Here $b_{\parallel} = b_x$ on an x -directed interval and $b_{\parallel} = b_y$ on a y -directed interval.